INCH-POUND

A-A-51151C
18 August 2005
SUPERSEDING
A-A-51151B
5 September 1996

COMMERCIAL ITEM DESCRIPTION

FILES, ROTARY, TREE SHAPE, RADIUS NOSE, HIGH SPEED STEEL OR TUNGSTEN CARBIDE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for tree shape, radius nose, high speed steel or tungsten carbide type rotary files intended for general-purpose machining and deburring.

2. CLASSIFICATION. The rotary files shall be of the following types, styles, sizes, and classes:

2.1 Type. The type to be furnished shall be as specified (see 7.3(b)).

Type I - High speed steel
Type II - Tungsten carbide

2.2 Style. The style to be furnished shall be as specified (see 7.3(c)).

Style A - Right-hand spiral fluting
Style B - Right-hand spiral fluting with chip breaker
Style C - Double cut fluting
Style D - Diamond cut fluting

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3455

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2.3 Size. The size to be furnished shall be as specified in table I for type I files or table II for type II files (see 7.3(d)). If a rotary file is to be furnished with a shank diameter or cutter head diameter and length different from those listed in tables I and II, the shank diameter and cutter head diameter and head length shall be specified in the acquisition order (see 7.3(e)).

TABLE I. Type I, high speed rotary files, sizes and characteristics.

Size	Characteristics		
	Shank diameter (inch)	Cutter head dimensions (inch)	
		Diameter	Head length
1	1/8	1/4	1/2
2	1/4	1/4	3/4
3	1/4	1/4	1-1/2
4	1/4	3/8	3/4
5	1/4	1/2	1
6	1/4	1/2	1-1/8
7	1/4	5/8	1-1/8
8	1/4	3/4	1-1/4
9	1/4	1-1/8	2

TABLE II. Type II, tungsten carbide rotary files, sizes and characteristics.

Size	Characteristics		
	Shank diameter (inch)	Cutter head dimensions (inch)	
		Diameter	Head length
1	1/8	1/8	1/4
2	1/8	1/8	1/2
3	1/8	1/4	1/2
4	1/4	1/4	5/8
5	1/4	3/8	3/4
6	1/4	1/2	1
7	1/4	5/8	1
8	1/4	3/4	1
9	1/4	3/4	1-1/4
10	1/4	3/4	1-1/2

2.4 Class. The class of rotary file to be furnished shall be as listed in table III for type I and table IV for type II files (see 7.3(f)).

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TABLE III. Number of teeth per circumference, type I rotary files.

Cutting head diameter (in.)	Teeth per circumference			
	Class A coarse cut	Class B medium cut	Class C fine cut	Class D extra coarse cut
1/8	6 - 8	9 - 11	12 - 15	4 - 5
3/16	9 - 11	12 - 14	15 - 20	4 - 8
1/4	10 - 13	14 - 16	17 - 21	4 - 9
5/16	11 - 16	17 - 20	21 - 26	5 - 10
3/8	12 - 17	18 - 22	23 - 30	5 - 11
7/16	13 - 18	19 - 23	24 - 31	6 - 12
1/2	14 - 20	21 - 26	27 - 35	7 - 13
5/8	17 - 22	23 - 29	30 - 40	9 - 16
3/4	18 - 24	25 - 32	33 - 44	10 - 17
7/8	19 - 25	26 - 33	34 - 45	13 - 18
1	20 - 27	28 - 41	42 - 55	15 - 19

TABLE IV. Number of teeth per circumference, type II rotary files.

Cutting head diameter (in.)	Teeth per circumference			
	Class A coarse cut	Class B medium cut	Class C fine cut	Class D extra coarse cut
1/8	8 - 11	12 - 15	16 - 20	4 - 5
3/16	10 - 12	13 - 16	17 - 24	4 - 8
1/4	12 - 15	16 - 20	21 - 26	4 - 9
5/16	13 - 16	17 - 22	23 - 30	5 - 10
3/8	13 - 18	19 - 26	27 - 38	5 - 11
7/16	15 - 20	21 - 27	28 - 36	6 - 12
1/2	16 - 22	23 - 28	29 - 39	7 - 13
5/8	18 - 25	26 - 38	39 - 47	9 - 16
3/4	19 - 28	29 - 38	39 - 49	10 - 17
7/8	22 - 29	30 - 37	38 - 55	13 - 18
1	24 - 32	33 - 43	44 - 62	15 - 19

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3. SALIENT CHARACTERISTICS

3.1 General requirements.

3.1.1 Type I rotary files. The cutting heads of type I rotary files shall meet the minimum requirements for standard high-speed steel described in ASTM A 600, "Standard Specification for Tool Steel High Speed", when tested in accordance with ASTM A 370, "Standard Test Methods and Definitions for Mechanical Testing of Steel Products".

3.1.2 Type II rotary files. The cutting heads of type II rotary files shall be manufactured from tungsten carbide with a hardness of not less than 91.2 and not greater than 92.5 on the Rockwell "A" scale when tested in accordance with ASTM B 294, "Standard Test Method for Hardness Testing of Cemented Carbides". The cutting heads of type II rotary files shall have a density of 14.6 to 15.1 grams per cubic centimeter when tested in accordance with ASTM B 311, "Test Method for Density Determination for Powder Metallurgy (P/M) Materials Containing Less Than Two Percent Porosity". Type II rotary files shall have a transverse rupture strength of 220,000 pounds per square inch when tested in accordance with ASTM B 406, "Standard Test Method for Transverse Rupture Strength of Cemented Carbides".

3.2 Fluting. The fluting of the rotary files shall be either right-hand spiral, right-hand spiral with chip breaker, double cut, or diamond cut, depending upon the style specified.

3.2.1 Style A rotary files. Style A rotary files shall have right-hand spiral fluting consisting of a right-hand helix. The helix angle measured from the centerline of the cutter head shall be between 15 degrees and 35 degrees.

3.2.2 Style B rotary files. Style B rotary files shall have right-hand spiral fluting with a left-hand spiral chip breaker along the periphery of the cutting teeth.

3.2.3 Style C rotary files. Style C rotary files shall have double cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals that results in a wedge or chisel form cutting tooth. The number of left-hand spirals shall be from 40 to 60 percent of the number of right-hand spirals.

3.2.4 Style D rotary files. Style D rotary files shall have diamond cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals that result in a pyramidal form cutting tooth. The number of left-hand spirals shall be from 70 to 90 percent of the number of right-hand spirals.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

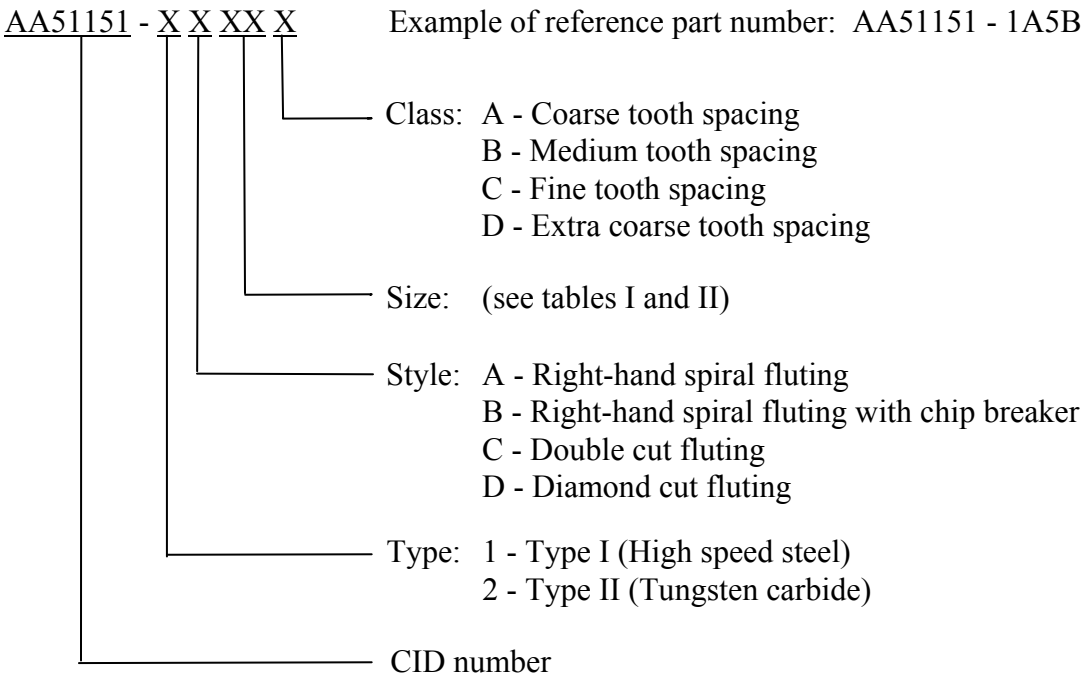
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(g)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



AA51151 - 1A5B identifies a high speed steel rotary file with right hand spiral fluting, a shank diameter of 1/4 inch, a cutter head 1 inch long with a diameter of 1/2 inch, with 21 to 26 teeth per circumference.

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7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

7.3 Ordering data. The acquisition order should specify the following:

a. CID document number, revision, and CID PIN.
b. Type of rotary files (see 2.1).
c. Style of rotary files (see 2.2).
d. Size of rotary files (see 2.3).
e. Shank and cutter head diameters and length, if different (see 2.3).
f. Class of rotary files (see 2.4).
g. Preservation, packing, and marking (see 6.1).

7.4 Subject term (key word) listing.

cutting head
deburring
fluting
shank diameter
teeth

MILITARY INTERESTS:

Custodian:
Air Force - 99

Review Activity:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS6

(Project 3455-0184)

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